LS-UQ Schema Extension Proposal

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1. Executive Summary

This document proposes extensions to the **LS-UQ (Layered Standard for Uncertainty Quantification) XML schema** to support the concepts defined in the OpenSCALING WP2/WP5 mapping workshop and the LS-UQ workflow composition document.

The current LS-UQ schema (`UncertaintyQuantification.xsd`) is designed around a single activity: **Forward Uncertainty Quantification**. It handles model information, uncertain parameters, operational domains, sampling methods, simulation settings, and desired results. However, the Credible Process Framework for Simulation requires broader coverage.

Gaps identified in the current schema:

- **Domain hierarchy** distinguishing ODD, OD, and Model OD
- **Activity-specific domains** for Verification, Calibration, Validation, Sensitivity Analysis, and Forward UQ
- **Credibility levels** tied to simulation purpose (framework-agnostic; OpenSCALING CL1-CL4 as default)
- **Domain coverage/gap assessment** between requested and realized domains
- **Process context** references to Credible Decision Process (CDP), Credible Simulation Process (CSP), and Credible Modeling Process (CMP) phases
- **Assumptions framework** distinguishing **required assumptions** from **modeling assumptions**

What this proposal adds:

Two new schema files (`Assumptions.xsd` and `DomainHierarchy.xsd`) and extensions to the main schema.

Not addressed in this version: binding assumptions and parameters to simulation model variables (e.g., mapping assumptions to FMU variables). This will be handled in a future version.

1.1. Schema Overview

For a visual overview of the full schema structure, see Schema Overview Diagram.

2. Abbreviations

Abbreviation	Full Name
CDP	Credible Decision Process
CL	Credibility Level
CMP	Credible Modeling Process
CSP	Credible Simulation Process
DTMD	Decision Task Meta Data
FMU	Functional Mock-up Unit
LS-UQ	Layered Standard for Uncertainty Quantification
M&S	Modeling and Simulation
OD	Operational Domain (= Working Range)
ODD	Operational Design Domain
SA	Sensitivity Analysis
SRMD	Simulation Resource Meta Data
SSC	System Structure Common (SSP schema)
SSP	System Structure and Parameterization
SSV	System Structure Parameter Values (SSP schema)
STC	SSP Traceability Common
STMD	Simulation Task Meta Data
UQ	Uncertainty Quantification
V&V	Verification and Validation

3. Background

3.1. The Credible Process Framework

Process	Full Name	Key Artifacts
CDP	Credible Decision Process	Simulation Request, Simulation Delivery
CSP	Credible Simulation Process	Modeling Request, Modeling Delivery
CMP	Credible Modeling Process	OD, Model OD, all activity domains (Do-domains)

The **Product Development Process** sits above these, with the **Operational Design Domain (ODD)** defined at the product level.

3.2. Modeling and Simulation (M&S) Credibility Levels

The schema represents credibility levels and simulation purposes as `xs:string`, not fixed enumerations. Organizations can plug in their own credibility framework (e.g., ASME V&V, NASA-STD-7009) by setting the `standard` attribute on `CredibilityAssessment`. See Design Decision #8.

Higher credibility levels impose stricter requirements on domain definitions, activity coverage, and assumption documentation.

Table 1. OpenSCALING recommended defaults (CL1-CL4)

Level	Purpose	Description
CL1	Trend behavior	Design support by relative comparisons or early concept evaluation. Not applicable for product Verification and Validation (V&V).
CL2	Quantitative predictions	Design support by quantitative predictions, prepare physical V&V.
CL3	Design evaluation	Reliable quantitative predictions with reduced number of physical tests. Requires statistical proof for the main working range.
CL4	Virtual Product V&V	Without additional physical test of corresponding requirements. Requires full working range UQ, data management, reproducibility, and traceability. M&S results are comparable with and can replace physical tests.

3.3. Current Schema Scope

The current schema (`UncertaintyQuantification.xsd` + `UncertainParameterType.xsd`) covers:

- `UncertaintyQuantification` root element with `Model` and `Activities`
- `ForwardUncertaintyQuantificationType` with ParameterSet, ObservedVariables, DesiredResults, SimulationSetting, SamplingMethod
- `OperationalDomainType` with Axes, Boundary (ConvexHull / HyperRectangle), ExperimentPoints
- `UncertainParameterType` with distribution types (Normal, Uniform, Weibull, Cauchy, monotone spline CDF, vendor-specific) and optional tolerance variants
- System Structure and Parameterization (SSP) interoperability via SSC/SSV imports

These capabilities are preserved unchanged.

3.4. SSP Traceability Integration

LS-UQ imports `STC.xsd` (SSP Traceability Common) and reuses its types directly for SSP Traceability compatibility. See Section 9 for architecture diagrams, integration points, usage scenarios, and upgrade paths.

4. Design Decisions

#	Decision	Rationale
1	Separate `.xsd` files for Assumptions and DomainHierarchy	Keeps the main schema readable. Allows independent versioning. Each file is self-contained in purpose.
2	All new elements are optional	Existing documents remain valid without modification.
3	Domain hierarchy via `parentRef` references, not XML nesting	Allows flexible relationships: multiple Model ODs can reference the same OD. Avoids deeply nested XML. Consistent with how SSP handles cross-references.
4	Activity types as separate elements (not a generic "Activity" with a `kind` attribute)	Type safety. Each activity can have unique children (e.g., `CalibrationTargets`, `ErrorMetrics`). XML schema validation catches structural errors at parse time.
5	Reuse existing geometry types (AxesType, BoundaryType, ExperimentPointsType)	No duplication. TypedOperationalDomain and ActivityDomain use the same boundary definitions as the existing OperationalDomain.
6	RequiredAssumptions and ModelingAssumptions share `AssumptionType`	The structure of an assumption is identical regardless of phase. The distinction is purely positional (before Model vs. inside Model).
7	`DomainCoverage` captures results, not algorithms	Geometric computation (intersection, gap analysis) is tool-specific. The schema captures the **outcome** (Covered, HighRisk, etc.) for exchange and documentation.
8	Credibility levels and simulation purposes use `xs:string` instead of fixed enumerations	Flexibility to support different credibility frameworks (OpenSCALING CL1-CL4, ASME V&V, NASA-STD-7009, etc.). Recommended values are documented but not enforced. Aligns with SSP Traceability's Classification approach. The `standard` attribute on `CredibilityAssessment` identifies which framework is used.
9	LS-UQ imports STC.xsd and references the `stc:GElementCommon` group (providing `Classification` and `Annotations`) rather than redefining equivalent types	Avoids duplication. Ensures LS-UQ data is directly compatible with SSP Traceability tooling. LS-UQ works standalone because importing STC types does not require adopting the full GlueParticle workflow.

#	Decision	Rationale
10	`ProcessContextType` uses `xs:string` + `glueParticleRef` instead of LS-UQ-defined phase enums	Process phases are authoritatively defined in STMD.xsd/DTMD.xsd. LS-UQ should not redefine them. String attributes allow standalone use; `glueParticleRef` enables direct linking when SSP Traceability is adopted.
11	Root element renamed `ExperimentSetup` → `UncertaintyQuantification`; child `ExperimentSet` → `Activities`; type `ExperimentSetType` → `ActivitiesType`	The original names were inherited from early prototyping. `UncertaintyQuantification` accurately describes the document's scope. `Activities` reflects that the container holds multiple activity types, not just experiment sets. The standard is not yet in production use, so this is a clean rename with no migration concerns.
12	Boundary types reduced to `ConvexHull` and `HyperRectangle`; `Box` and `UnitHyperCube` removed	`Box` is ambiguous. `HyperRectangle` is mathematically correct and unambiguous for an axis-aligned bounding box. It uses exactly two `Point` elements (min-corner and max-corner), each with a `coordinates` attribute listing values in axis order — consistent with `ConvexHull`'s `Point` syntax. `UnitHyperCube` is a special case of `HyperRectangle` (all axes 0-1) and does not warrant its own element.
13	Activity element order: Verification, Calibration, Validation, SA, Forward UQ	Matches the logical workflow dependency chain (ASME V&V 20, NASA-STD-7009B). Document order only; no schema enforcement.
14	Domain kind uses a fixed `TypedOperationalDomainKind` enum instead of `xs:string`	The three domain kinds (OperationalDesignDomain - ODD, RequestedOperationalDomain - OD, ModeledOperationalDomain - Model OD) are structural to LS-UQ's ODD → OD → Model OD hierarchy and do not vary across organizations. Unlike credibility levels, these are defined by the standard itself. The former `CL4OD` value is removed — credibility qualification belongs on `CredibilityAssessment`, not the domain kind.

#	Decision	Rationale
15	Reusable `AExternalSource` attribute group for externalizing large data	Elements like `Parameters`, `Axes`, `ConvexHull`, `HyperRectangle`, `Points`, `Units`, `ObservedVariables`, `DesiredResults`, `SimulationSetting`, and `SamplingMethod` can grow very large or reference external configuration files. Following SSP/FMI conventions (`source` + `type` + `sourceBase` + integrity checksum), a single reusable attribute group is applied to all ten types. When `source` is present, inline children may be omitted.

5. Proposed File Structure

The proposal introduces two new schema files and modifies the main schema. The `UncertainParameterType.xsd` is **unchanged**. The schema imports `STC.xsd` (SSP Traceability Common) as an external dependency; it must be available at schema resolution time but is not part of the LS-UQ file set.

```
schema/
 ├─── current/                              ①
 │    ├─── UncertaintyQuantification.xsd
 │    └─── UncertainParameterType.xsd
 └─── proposed/                             ②
      ├─── UncertaintyQuantification.xsd    (MODIFIED)
      ├─── UncertainParameterType.xsd       (UNCHANGED)
      ├─── Assumptions.xsd                  (NEW)
      └─── DomainHierarchy.xsd             (NEW)

examples/
 ├─── example_instance.xml                  ③
 ├─── example_assumptions_only.xml          ④
 ├─── example_domains_only.xml              ⑤
 ├─── example_activities.xml                ⑥
 ├─── example_external_source.xml           ⑦
```

① Original schema files for reference

② Proposed schema files

③ Complete example exercising all features

④ Focused example: assumptions only

⑤ Focused example: domain hierarchy only

⑥ Focused example: multiple activity types

⑦ Focused example: external source attributes (inline vs external data)

The main schema includes the new files via `xs:include`:

```
<xs:include schemaLocation="UncertainParameterType.xsd" />
<xs:include schemaLocation="Assumptions.xsd" />           <!-- [NEW] -->
<xs:include schemaLocation="DomainHierarchy.xsd" />       <!-- [NEW] -->
```

6. Extension 1: Assumptions Framework

6.1. Motivation

The LS-UQ workflow document identifies a distinction between two phases of assumptions:

Before UncertaintyQuantification, RequiredAssumptions

Assumptions defined as requirements or constraints for modelers. Part of the specification and planning phase. Stated as: *"The model **shall** fulfill these assumptions."*

After UncertaintyQuantification (inside Model), ModelingAssumptions

Assumptions documented as characteristics of what was actually built. Part of the model metadata. Stated as: *"The model **does** fulfill these assumptions."*

This distinction enables comparing what was **required** versus what was **realized**.

6.2. Schema: Assumptions.xsd

`Assumptions.xsd` defines the following types:

6.2.1. OriginType

Where an assumption came from:

```
<xs:simpleType name="OriginType">
  <xs:restriction base="xs:string">
    <xs:enumeration value="requirement"/>
    <xs:enumeration value="standard"/>
    <xs:enumeration value="design-decision"/>
    <xs:enumeration value="inherited"/>
    <xs:enumeration value="expert-judgment"/>
    <xs:enumeration value="empirical"/>
    <xs:enumeration value="literature"/>
    <xs:enumeration value="unknown"/>
  </xs:restriction>
</xs:simpleType>
```

6.2.2. AssumptionType

A single assumption. Attributes and children:

- **Identity:** `id`, `name`, `description`
- **Provenance:** `origin` (OriginType enum), `originRef` (free text reference)
- **Time-dependent validity:** `validFrom`, `validTo`, `timeUnit`
- **Traceability:** via `ssc:ATopLevelMetaData` (SSP), providing `author`, `generationDateAndTime`,

generationTool, fileversion, copyright, license

- **Value:** one of several typed child elements (see Section 6.2.3)
- **OperationalDomainRef:** optional link to the domain where this assumption applies

```
<xs:complexType name="AssumptionType">
  <xs:sequence>
    <!-- Optional: link to an operational domain this assumption applies to -->
    <xs:element name="OperationalDomainRef" type="uq:OperationalDomainRefType"
                minOccurs="0" maxOccurs="unbounded"/>
    <!-- The assumption value: at most one of these -->
    <xs:choice minOccurs="0">
      <xs:element name="Real" type="uq:RealValueType"/>
      <xs:element name="RealRange" type="uq:RealRangeType"/>
      <xs:element name="String" type="uq:StringValueType"/>
      <xs:element name="Text" type="xs:string"/>
      <xs:element name="ExternalDocument" type="uq:ExternalDocumentType"/>
    </xs:choice>
    <!-- STC common elements: Classification (0..*) + Annotations (0..1) -->
    <xs:group ref="stc:GElementCommon"/>
  </xs:sequence>
  <!-- Identity -->
  <xs:attribute name="id" type="xs:ID" use="optional"/>
  <xs:attribute name="name" type="xs:string" use="required"/>
  <xs:attribute name="description" type="xs:string" use="optional"/>
  <!-- Provenance -->
  <xs:attribute name="origin" type="uq:OriginType" use="optional"/>
  <xs:attribute name="originRef" type="xs:string" use="optional"/>
  <!-- Time-dependent validity (simulation time) -->
  <xs:attribute name="validFrom" type="xs:double" use="optional"/>
  <xs:attribute name="validTo" type="xs:double" use="optional"/>
  <xs:attribute name="timeUnit" type="xs:string" use="optional"/>
  <!-- Traceability (reuses SSP/SSC) -->
  <xs:attributeGroup ref="ssc:ATopLevelMetaData"/>
</xs:complexType>
```

6.2.3. Assumption Value Types

Each assumption can carry at most one typed value:

Type	Attributes	Use Case
Real	value, unit	Single numeric value (e.g., atmospheric pressure = 101325 Pa)
RealRange	low, high, unit	Valid range (e.g., operating temperature 25-40°C)
String	value	Short categorical value (e.g., terrain = "flat")

Type	Attributes	Use Case
`Text`	*(text content)*	Long-form description or justification
`ExternalDocument`	`source`, `type`, `section`, `description`	Reference to external file or URL

6.2.4. RequiredAssumptionsType and ModelingAssumptionsType

Both are containers for `Assumption` elements with audit metadata:

```
<!-- RequiredAssumptionsType — pre-experiment requirements                -->
<!-- "The model SHALL assume these conditions"                            -->
<!-- ===================================================================== -->
<xs:complexType name="RequiredAssumptionsType">
  <xs:sequence>
    <xs:element name="Assumption" type="uq:AssumptionType"
                minOccurs="1" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:attribute name="id" type="xs:ID" use="required"/>
  <xs:attribute name="description" type="xs:string" use="optional"/>
  <xs:attributeGroup ref="ssc:ATopLevelMetaData"/>
</xs:complexType>
```

`ModelingAssumptionsType` has the same structure. Both use `ssc:ATopLevelMetaData` for traceability (author, generationDateAndTime, etc.).

6.3. Where Assumptions Sit in UncertaintyQuantification

```
<UncertaintyQuantification name="..."
    requiredAssumptionsRefs="thermal_req safety_req">   ①

  <RequiredAssumptions id="thermal_req" ...>             ②
    <Assumption .../>
  </RequiredAssumptions>

  <Domains>...</Domains>

  <Model name="..." file="...">
    <ModelingAssumptions id="ma_vehicle_v1" ...>          ③
      <Assumption .../>
    </ModelingAssumptions>
  </Model>

  <Activities>...</Activities>
</UncertaintyQuantification>
```

① UncertaintyQuantification references required assumptions by ID

② RequiredAssumptions appear **before** the Model (specification phase)

③ ModelingAssumptions live **inside** the Model (realization phase)

6.4. Example: Assumptions in Practice

Example with all assumption value types, time-dependent validity, and the required-vs-modeling distinction:

```
<?xml version="1.0" encoding="UTF-8"?>
<!-- Example: Assumptions feature only (minimal) -->
<uq:UncertaintyQuantification
    xmlns:uq="http://openscaling.org/UQ1/UncertaintyQuantification"
    name="AssumptionsDemo"
    requiredAssumptionsRefs="thermal_req_v1">

  <!-- Required assumptions: what the model SHALL assume -->
  <uq:RequiredAssumptions id="thermal_req_v1"
      generationDateAndTime="2024-03-01T00:00:00"
      author="safety@company.com">

    <uq:Assumption id="ra_temp_01"
        name="operatingTemperature"
        description="Valid operating temperature range"
        origin="requirement">
      <uq:RealRange low="25" high="40" unit="degC"/>
    </uq:Assumption>

    <uq:Assumption id="ra_press_01"
        name="ambientPressure"
        origin="standard"
        originRef="ISO2533">
      <uq:Real value="101325" unit="Pa"/>
    </uq:Assumption>

  </uq:RequiredAssumptions>

  <uq:Model name="VehicleThermal" file="vehicle.fmu">

    <!-- Modeling assumptions: what the model DOES assume -->
    <uq:ModelingAssumptions id="ma_vehicle_v1"
        generationDateAndTime="2025-12-10T00:00:00"
        author="amin.bajand@liu.se">

      <!-- Single real value -->
      <uq:Assumption id="ma_01"
          name="ambientPressure"
          origin="standard"
          originRef="ISO2533">
```

```
        <uq:Real value="101325" unit="Pa"/>
      </uq:Assumption>

      <!-- Range with time-dependent validity -->
      <uq:Assumption id="ma_02"
          name="operatingTemperature"
          origin="requirement"
          originRef="ra_temp_01"
          validFrom="100"
          timeUnit="s">
        <uq:RealRange low="-10" high="80" unit="degC"/>
      </uq:Assumption>

      <!-- Free text assumption -->
      <uq:Assumption id="ma_03"
          name="linearCooling"
          origin="empirical"
          originRef="test_report_2024.pdf">
        <!-- References a domain defined in the full instance (see
example_instance.xml) -->
        <uq:OperationalDomainRef ref="od_normal_operation"/>
        <uq:Text>Cooling follows Newton's law with constant heat transfer
coefficient</uq:Text>
      </uq:Assumption>

      <!-- Time-dependent assumption -->
      <uq:Assumption id="ma_04"
          name="coldStartBehavior"
          origin="design-decision"
          validFrom="0"
          validTo="100"
          timeUnit="s">
        <uq:Text>Engine temperature interpolates linearly from ambient to
90C</uq:Text>
      </uq:Assumption>

      <!-- External document reference -->
      <uq:Assumption id="ma_05"
          name="materialProperties"
          origin="literature"
          originRef="ASM Handbook">
        <uq:ExternalDocument
            source="materials/steel_properties.pdf"
            type="application/pdf"
            section="Chapter 3.2"
            description="Material property tables"/>
      </uq:Assumption>

      <!-- Flow rate assumption: model assumes unidirectional coolant flow.
           Near zero flow, pressure fluctuations can cause flow reversal,
           leading to solver instability (upstream/downstream port switching). -->
```

```
      <uq:Assumption id="ma_06"
          name="coolantFlowRate"
          origin="design-decision"
          description="Unidirectional flow; not valid near zero flow">
        <uq:RealRange low="0.5" high="3.0" unit="kg/s"/>
      </uq:Assumption>

    </uq:ModelingAssumptions>

  </uq:Model>

  <uq:Activities id="es_01">
    <uq:ForwardUncertaintyQuantification id="fuq_01">
      <!-- ... existing ForwardUQ content ... -->
    </uq:ForwardUncertaintyQuantification>
  </uq:Activities>

</uq:UncertaintyQuantification>
```

In this example:

- `RequiredAssumptions` defines requirements (temperature in the range [25, 40] [°C]); `ModelingAssumptions` documents the realized model (temperature in the range [-10, 80] [°C], wider than required)
- Assumption `ma_04` uses time-dependent validity (valid only during cold start: [0, 100] [s])
- Assumption `ma_05` references an external PDF; `ma_03` links to an operational domain
- Assumption `ma_06` constrains coolant flow rate to [0.5, 3.0] [kg/s], preventing use in engine-off or pump-failure scenarios

7. Extension 2: Domain Hierarchy

7.1. Motivation

The Credible Process Framework uses multiple domains at different levels:

Domain	Short Name	Description
ODD	Operational Design Domain	Product-level scope. The broadest intended operating envelope. Defined at the product architecture level.
OD	Requested Operational Domain (= Working Range)	The requested input space where the model is to be used. Defined in the Simulation Request / Modeling Request. Corresponds to the sought purpose of the model.
Model OD	Modeled Operational Domain	What the modeler **actually realized**. May be a subset of OD if the modeler could not cover the full requested range.

These domains form an intended hierarchy: the **ODD** (`OperationalDesignDomain`) is the broadest scope, the **OD** (`RequestedOperationalDomain`) is a subset of the ODD, and the **Model OD** (`ModeledOperationalDomain`) is intended to cover the OD but may in practice be smaller (not all requested regions realized) or extend beyond it in some dimensions (additional regions not requested). Regions of the Model OD that fall outside the OD are classified as **NotNeeded** in the domain coverage assessment. A CL4 V&V domain is expressed as a `ModeledOperationalDomain` with an appropriate `CredibilityAssessment`.

Additionally, each credibility activity (Verification, Calibration, Validation, Sensitivity Analysis, Forward UQ) operates on its own sub-domain of the Model OD:

- **DoVerification** = Domain of Verification
- **DoCalibration** = Domain of Calibration
- **DoV (DoValidation)** = Domain of Validation
- **DoSA** = Domain of Sensitivity Analysis
- **DoUQ** = Domain of Uncertainty Quantification

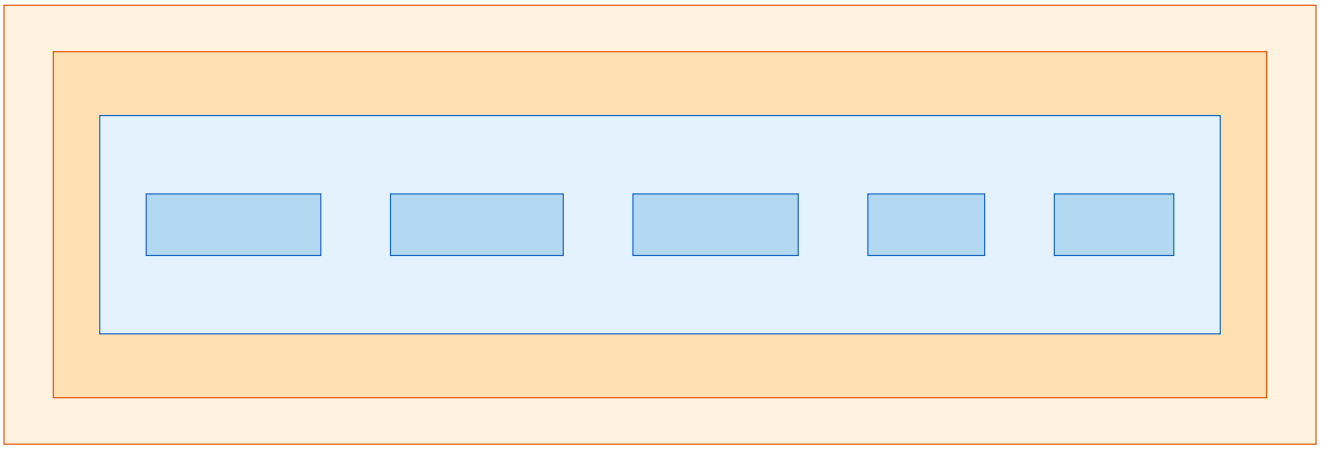


Figure 1. Domain hierarchy: ODD contains OD, which contains Model OD and its activity sub-domains

7.2. Schema: DomainHierarchy.xsd

7.2.1. TypedOperationalDomainType

Wraps the existing `OperationalDomainType` geometry with metadata:

```
  <xs:complexType name="TypedOperationalDomainType">
    <xs:sequence>
      <!-- The geometric definition — reuses the existing structure -->
      <xs:element name="Axes" type="uq:AxesType" minOccurs="0"/>
      <xs:element name="Boundary" type="uq:BoundaryType" minOccurs="0"/>
      <xs:element name="ExperimentPoints" type="uq:ExperimentPointsType" minOccurs=
"0"/>
      <!-- Credibility assessment -->
      <xs:element name="CredibilityAssessment" type="uq:CredibilityAssessmentType"
                  minOccurs="0"/>
      <!-- Process context -->
      <xs:element name="ProcessContext" type="uq:ProcessContextType"
                  minOccurs="0"/>
      <!-- Free-text description -->
      <xs:element name="Description" type="xs:string" minOccurs="0"/>
      <!-- STC common elements: Classification (0..*) + Annotations (0..1) -->
      <xs:group ref="stc:GElementCommon"/>
    </xs:sequence>
    <!-- Identity -->
    <xs:attribute name="id" type="xs:ID" use="required"/>
    <xs:attribute name="name" type="xs:string" use="optional"/>
    <!-- Domain kind: ODD, Requested OD, or Modeled OD -->
    <xs:attribute name="kind" type="uq:TypedOperationalDomainKind" use="required"/>
    <xs:attribute name="geometryKind" type="xs:string" use="optional"/>
    <!-- Hierarchy: reference to parent domain -->
    <xs:attribute name="parentRef" type="xs:string" use="optional"/>
    <xs:attribute name="srmdRef" type="xs:anyURI" use="optional"/>
  </xs:complexType>
```

The hierarchy uses `parentRef` references rather than XML nesting (see Design Decision #3).

7.2.2. ActivityDomainType

Each activity domain is a sub-region of a parent domain (typically Model OD):

```
  <!-- ActivityDomainType — a domain scoped to a specific activity         -->
  <!-- ================================================================== -->
  <!--
    From the presentation: DoVerification, DoCalibration, DoV, DoSA, DoUQ
    are each sub-regions of the ModelOD with their own boundary.
    Each activity domain carries:
```

```
      - Which activity it serves (kind)
      - Its own geometric boundary (may be a subset of the parent ModelOD)
      - Reference to the parent typed domain (typically ModelOD)
      - Optional settings and results specific to the activity
  -->
  <xs:complexType name="ActivityDomainType">
    <xs:sequence>
      <!-- Geometric boundary of this activity domain -->
      <xs:element name="Axes" type="uq:AxesType" minOccurs="0"/>
      <xs:element name="Boundary" type="uq:BoundaryType" minOccurs="0"/>
      <xs:element name="ExperimentPoints" type="uq:ExperimentPointsType" minOccurs=
"0"/>
      <!-- Activity-specific settings (forward propagation config, etc.) -->
      <xs:element name="Settings" type="uq:ActivitySettingsType" minOccurs="0"/>
      <!-- Activity-specific results -->
      <xs:element name="ResultSet" type="uq:ResultSetType" minOccurs="0"/>
      <!-- Free-text description -->
      <xs:element name="Description" type="xs:string" minOccurs="0"/>
      <!-- STC common elements: Classification (0..*) + Annotations (0..1) -->
      <xs:group ref="stc:GElementCommon"/>
    </xs:sequence>
    <!-- Identity -->
    <xs:attribute name="id" type="xs:ID" use="required"/>
    <xs:attribute name="name" type="xs:string" use="optional"/>
    <!-- What kind of activity -->
    <xs:attribute name="kind" type="xs:string" use="required"/>
    <!-- Reference to parent domain (typically a TypedOperationalDomain) -->
    <xs:attribute name="domainRef" type="xs:string" use="optional"/>
    <!-- URI reference to an SRMD file describing this domain -->
    <xs:attribute name="srmdRef" type="xs:anyURI" use="optional"/>
  </xs:complexType>
```

Activity kinds use `xs:string` with recommended values: Verification, Calibration, Validation, UncertaintyQuantification, SensitivityAnalysis.

7.2.3. DomainCoverageType (Gap Analysis)

The gap between the **requested** OD and the **realized** Model OD has risk implications:

Classification	Color (in PPTX)	Meaning
Covered	Green	Model operates correctly. Decision-making is supported.
HighRisk	Red	Critical gap. Model may execute but results are unreliable. High risk for model-based decisions.
AcceptableRisk	Orange	Gap exists but risk is potentially acceptable. Limited data or extrapolation.
NotNeeded	(outside OD)	Region outside the requested OD. No model coverage required.

```
<xs:complexType name="DomainCoverageType">
  <xs:sequence>
    <xs:element name="Region" minOccurs="0" maxOccurs="unbounded">
      <xs:complexType>
        <xs:sequence>
          <xs:element name="Boundary" type="uq:BoundaryType" minOccurs="0"/>
          <xs:element name="Description" type="xs:string" minOccurs="0"/>
        </xs:sequence>
        <xs:attribute name="classification" type="uq:CoverageClassificationType"
                      use="required"/>
        <xs:attribute name="id" type="xs:ID" use="optional"/>
      </xs:complexType>
    </xs:element>
    <xs:element name="Description" type="xs:string" minOccurs="0"/>
  </xs:sequence>
  <xs:attribute name="id" type="xs:ID" use="optional"/>
  <!-- The two domains being compared -->
  <xs:attribute name="requestedDomainRef" type="xs:string" use="required"/>
  <xs:attribute name="realizedDomainRef" type="xs:string" use="required"/>
</xs:complexType>
```

7.2.4. CredibilityAssessmentType and ProcessContextType

```
<xs:complexType name="CredibilityAssessmentType">
  <xs:sequence>
    <xs:element name="Description" type="xs:string" minOccurs="0"/>
    <!-- STC common elements: Classification (0..*) + Annotations (0..1) -->
    <xs:group ref="stc:GElementCommon"/>
  </xs:sequence>
  <xs:attribute name="level" type="xs:string" use="required"/>
  <xs:attribute name="purpose" type="xs:string" use="optional"/>
  <!-- Indicates which credibility framework is used
       (e.g., "OpenSCALING", "ASME-VVUQ", "NASA-STD-7009") -->
  <xs:attribute name="standard" type="xs:string" use="optional"/>
</xs:complexType>

<!-- Process context: which process and phase an artifact belongs to -->
<xs:complexType name="ProcessContextType">
  <xs:sequence>
    <xs:element name="Description" type="xs:string" minOccurs="0"/>
    <!-- STC common elements: Classification (0..*) + Annotations (0..1) -->
    <xs:group ref="stc:GElementCommon"/>
  </xs:sequence>
  <!-- Simple string attributes for lightweight usage (standalone) -->
  <xs:attribute name="process" type="xs:string" use="optional"/>
  <xs:attribute name="phase" type="xs:string" use="optional"/>
  <xs:attribute name="description" type="xs:string" use="optional"/>
  <!-- URI reference to an STMD/DTMD GlueParticle (SSP Traceability) -->
  <xs:attribute name="glueParticleRef" type="xs:anyURI" use="optional"/>
```

```
  </xs:complexType>
```

7.2.5. DomainsType (Container)

All domain definitions go into one top-level container:

```
  <!-- DomainsType — top-level container for all domain definitions        -->
  <!-- ================================================================== -->
  <xs:complexType name="DomainsType">
    <xs:sequence>
      <xs:element name="TypedOperationalDomain" type="uq:TypedOperationalDomainType"
                  minOccurs="0" maxOccurs="unbounded"/>
      <xs:element name="ActivityDomain" type="uq:ActivityDomainType"
                  minOccurs="0" maxOccurs="unbounded"/>
      <xs:element name="DomainCoverage" type="uq:DomainCoverageType"
                  minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>
```

7.3. Example: Domain Hierarchy in Practice

```
<?xml version="1.0" encoding="UTF-8"?>
<!-- Example: Domain hierarchy and activity domains (minimal) -->
<uq:UncertaintyQuantification
    xmlns:uq="http://openscaling.org/UQ1/UncertaintyQuantification"
    name="DomainHierarchyDemo">

  <uq:Domains>

    <!-- Level 1: Requested OD (from model requester) -->
    <uq:TypedOperationalDomain id="od_requested"
        name="Requested Working Range"
        kind="RequestedOperationalDomain"
        geometryKind="HyperRectangle">
      <uq:Axes>
        <uq:Axis name="temperature" unit="degC"/>
        <uq:Axis name="speed" unit="km/h"/>
      </uq:Axes>
      <uq:Boundary>
        <uq:HyperRectangle>
          <uq:Point coordinates="-20 0"/>
          <uq:Point coordinates="50 250"/>
        </uq:HyperRectangle>
      </uq:Boundary>
      <uq:CredibilityAssessment level="CL4" purpose="VirtualProductVV"/>
      <uq:ProcessContext process="CSP"
          phase="DefinitionOfSimulationRequirements"/>
      <!-- Alternative for SSP Traceability users:
```

```
        <uq:ProcessContext glueParticleRef="metadata/simulation_task.stmd"/> -->
  </uq:TypedOperationalDomain>

  <!-- Level 2: Modeled OD (what modeler built) -->
  <uq:TypedOperationalDomain id="od_model"
      name="Realized Model Domain"
      kind="ModeledOperationalDomain"
      geometryKind="ConvexHull"
      parentRef="od_requested">
    <uq:Axes>
      <uq:Axis name="temperature" unit="degC"/>
      <uq:Axis name="speed" unit="km/h"/>
    </uq:Axes>
    <uq:Boundary>
      <uq:ConvexHull>
        <uq:Point coordinates="-10 0"/>
        <uq:Point coordinates="-10 200"/>
        <uq:Point coordinates="45 200"/>
        <uq:Point coordinates="45 0"/>
      </uq:ConvexHull>
    </uq:Boundary>
    <uq:ProcessContext process="CSP" phase="ImplementModel"/>
  </uq:TypedOperationalDomain>

  <!-- Activity domain: DoVerification (subset of Model OD) -->
  <uq:ActivityDomain id="do_verif"
      name="Verification Domain"
      kind="Verification"
      domainRef="od_model">
    <uq:Axes>
      <uq:Axis name="temperature" unit="degC"/>
      <uq:Axis name="speed" unit="km/h"/>
    </uq:Axes>
    <uq:Boundary>
      <uq:HyperRectangle>
        <uq:Point coordinates="0 50"/>
        <uq:Point coordinates="30 150"/>
      </uq:HyperRectangle>
    </uq:Boundary>
    <uq:Settings method="GridConvergenceStudy"/>
  </uq:ActivityDomain>

  <!-- Activity domain: DoValidation -->
  <uq:ActivityDomain id="do_val"
      name="Validation Domain"
      kind="Validation"
      domainRef="od_model">
    <uq:Boundary>
      <uq:HyperRectangle>
        <uq:Point coordinates="-5 20"/>
        <uq:Point coordinates="40 180"/>
```

```
        </uq:HyperRectangle>
      </uq:Boundary>
      <uq:Settings method="ForwardUncertaintyPropagation"/>
    </uq:ActivityDomain>

    <!-- Activity domain: DoUQ -->
    <uq:ActivityDomain id="do_uq"
        name="UQ Domain"
        kind="UncertaintyQuantification"
        domainRef="od_model">
      <uq:Boundary>
        <uq:HyperRectangle>
          <uq:Point coordinates="-10 0"/>
          <uq:Point coordinates="45 200"/>
        </uq:HyperRectangle>
      </uq:Boundary>
      <uq:Settings method="ForwardUncertaintyPropagation"/>
    </uq:ActivityDomain>

    <!-- Gap analysis: requested vs realized -->
    <uq:DomainCoverage id="cov_01"
        requestedDomainRef="od_requested"
        realizedDomainRef="od_model">
      <uq:Region classification="Covered">
        <uq:Description>-10..45C x 0..200 km/h: model valid</uq:Description>
      </uq:Region>
      <uq:Region classification="HighRisk">
        <uq:Description>Below -10C: model unreliable</uq:Description>
      </uq:Region>
      <uq:Region classification="AcceptableRisk">
        <uq:Description>Above 200 km/h: limited data</uq:Description>
      </uq:Region>
    </uq:DomainCoverage>

  </uq:Domains>

  <uq:Model name="ThermalModel" file="model.fmu"/>

  <uq:Activities id="es_01" credibilityLevel="CL4">
    <uq:ForwardUncertaintyQuantification id="fuq_01"
        activityDomainRef="do_uq">
      <!-- ... existing ForwardUQ content ... -->
    </uq:ForwardUncertaintyQuantification>
  </uq:Activities>

</uq:UncertaintyQuantification>
```

This example shows:

- An **OD** (requested: -20..50°C x 0..250 km/h) and **Model OD** (realized: -10..45°C x 0..200 km/h)

linked via `parentRef`

- Three **activity domains** (DoVerification, DoValidation, DoUQ) referencing the Model OD
- A **DomainCoverage** with classified regions (Covered, HighRisk, AcceptableRisk)

8. Extension 3: Activity Types

8.1. Motivation

The current `ActivitiesType` only allows `ForwardUncertaintyQuantification`. The WP2/WP5 workshop requires additional peer-level activities:

1. **Verification**: confirms the model solves equations correctly (code verification, solution verification) and in a numerically robust way. Should logically precede Calibration.
2. **Calibration**: fits model parameters to observed data. Must precede Validation.
3. **Validation**: confirms the model represents reality (comparison with data taken as reference (e.g., experimental data) using calibrated parameters)
4. **Sensitivity Analysis**: identifies which inputs most influence outputs
5. **Forward UQ**: propagates uncertainties through the model *(existing)*

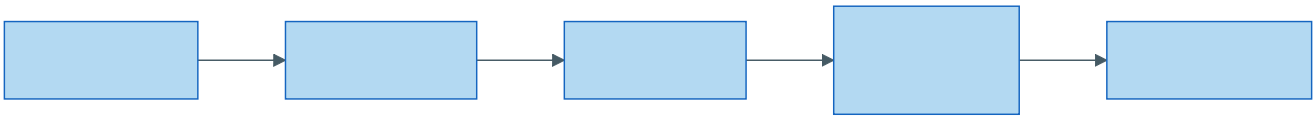


Figure 2. Recommended logical order of activities (document order only, not enforced by schema)

8.2. Changes to ActivitiesType

```
  <xs:complexType name="ActivitiesType">
    <!-- Activities are listed in recommended logical order:
         Verification → Calibration → Validation → SA → Forward UQ.
         This is document order only; the schema does not enforce execution order. -->
    <xs:sequence>
      <xs:element name="Verification" type="uq:VerificationType"
                  minOccurs="0" maxOccurs="unbounded"/>
      <xs:element name="Calibration" type="uq:CalibrationType"
                  minOccurs="0" maxOccurs="unbounded"/>
      <xs:element name="Validation" type="uq:ValidationType"
                  minOccurs="0" maxOccurs="unbounded"/>
      <xs:element name="SensitivityAnalysis" type="uq:SensitivityAnalysisType"
                  minOccurs="0" maxOccurs="unbounded"/>
      <xs:element name="ForwardUncertaintyQuantification"
        type="uq:ForwardUncertaintyQuantificationType"
        minOccurs="0" maxOccurs="unbounded"/>                <!-- [MODIFIED] was
required, now 0..* -->
      <!-- [NEW] STC common elements: Classification (0..*) + Annotations (0..1) -->
      <xs:group ref="stc:GElementCommon"/>
    </xs:sequence>
    <xs:attribute name="id" type="xs:ID" />
    <!-- [NEW] Optional credibility level for the entire activities container -->
    <xs:attribute name="credibilityLevel" type="xs:string" use="optional"/>
```

```
  </xs:complexType>
```

`ForwardUncertaintyQuantification` changes from required (1..1) to optional (0..*).

8.3. New Activity Types

8.3.1. VerificationType

Verification asks: *"Is the model solving the equations correctly?"*

It includes standard UQ elements (ParameterSet, SimulationSetting, etc.) plus verification-specific elements:

- `ReferenceData`: analytical solutions, benchmark data, or manufactured solutions
- `ErrorMetrics`: named metrics with values, thresholds, and pass/fail status (e.g., GCI, L2 norm, observed order of accuracy)

```
  <uq:Verification id="verif_01"
      name="Grid Convergence Study"
      activityDomainRef="do_verification">

    <uq:SimulationSetting stopTime="3600" interval="1.0"
        tolerance="1e-8" method="CVODE"/>

    <uq:ReferenceData description="Analytical solution">
      <uq:DataSource name="analytical_ref"
          file="reference/analytical_solution.csv"
          type="text/csv"/>
    </uq:ReferenceData>

    <uq:ResultSet>
      <uq:Results>
        <uq:ErrorMetric type="GCI" name="GCI_fine" value="0.012"
            threshold="0.05" pass="true"/>
        <uq:ErrorMetric type="observed_order" name="observed_order" value="1.95"
            threshold="1.8" pass="true"/>
      </uq:Results>
    </uq:ResultSet>

  </uq:Verification>
```

8.3.2. CalibrationType

Calibration fits model parameters to match observed data. It must run **before** Validation (you validate with calibrated parameters), but logically **after** Verification (there is no point calibrating if the model cannot solve equations correctly).

Unique element: `CalibrationTargets`, which defines the parameters to calibrate with initial values and bounds:

```
<uq:Calibration id="cal_01"
    name="Convection Coefficient Calibration"
    activityDomainRef="do_calibration">

  <uq:ParameterSet>
    <uq:Parameters>
      <uq:UncertainParameter name="h_convection" source="Estimated">
        <uq:Uniform minimum="10" maximum="50" unit="W/(m2.K)"/>
      </uq:UncertainParameter>
    </uq:Parameters>
  </uq:ParameterSet>

  <uq:SimulationSetting stopTime="1800" interval="1.0"
      tolerance="1e-6" method="CVODE"/>

  <uq:ReferenceData description="Lab measurements">
    <uq:DataSource name="lab_data_2024"
        file="reference/lab_convection.csv"
        type="text/csv"/>
  </uq:ReferenceData>

  <uq:CalibrationTargets>
    <uq:Target parameterName="h_convection"
        initialValue="25.0"
        lowerBound="10.0"
        upperBound="50.0"
        unit="W/(m2.K)"/>
  </uq:CalibrationTargets>

  <uq:ResultSet>
    <uq:Results>
      <uq:Normal mu="23.7" sigma="1.2"/>
    </uq:Results>
  </uq:ResultSet>

</uq:Calibration>
```

8.3.3. ValidationType

Validation compares simulation results against data taken as reference (e.g., physical test data). It shares the same structure as Verification:

```
<uq:Validation id="val_01"
    name="Wind Tunnel Comparison"
    activityDomainRef="do_validation">
```

```
      <uq:ParameterSet>
        <uq:Parameters>
          <uq:UncertainParameter name="T_ambient" source="Measured">
            <uq:Normal mu="22.0" sigma="1.0" unit="degC"/>
          </uq:UncertainParameter>
        </uq:Parameters>
      </uq:ParameterSet>

      <uq:SimulationSetting stopTime="1800" interval="0.5"
          tolerance="1e-6" method="CVODE"/>

      <uq:SamplingMethod>
        <uq:LatinHyperCube numberOfSamples="200"/>
      </uq:SamplingMethod>

      <uq:ReferenceData description="Wind tunnel data">
        <uq:DataSource name="wind_tunnel_2024"
            file="reference/wind_tunnel.csv"
            type="text/csv"/>
      </uq:ReferenceData>

      <uq:ResultSet>
        <uq:Results>
          <uq:ErrorMetric type="RMSE" name="RMSE" value="2.3"
              threshold="5.0" pass="true"/>
          <uq:ErrorMetric type="validationMetric" name="validation_metric"
value="0.87"
              threshold="0.80" pass="true"/>
        </uq:Results>
      </uq:ResultSet>

    </uq:Validation>
```

8.3.4. SensitivityAnalysisType

Reuses existing DesiredResults (particularly SobolIndices). Computed indices are stored in the `Results` element (of type `ActivityResultsType`) as `MetricValue` entries:

```
    <uq:SensitivityAnalysis id="sa_01"
        name="Global Sensitivity (Sobol)">

      <uq:ParameterSet>
        <uq:Parameters>
          <uq:UncertainParameter name="T_ambient" source="Measured">
            <uq:Uniform minimum="-10" maximum="45" unit="degC"/>
          </uq:UncertainParameter>
          <uq:UncertainParameter name="v_vehicle" source="Estimated">
            <uq:Uniform minimum="0" maximum="200" unit="km/h"/>
          </uq:UncertainParameter>
```

```
        <uq:UncertainParameter name="h_convection" source="Calibrated">
          <uq:Uniform minimum="15" maximum="35" unit="W/(m2.K)"/>
        </uq:UncertainParameter>
      </uq:Parameters>
    </uq:ParameterSet>

    <uq:ObservedVariables>
      <uq:ObservedVariable name="T_coolant_max"/>
    </uq:ObservedVariables>

    <uq:DesiredResults>
      <uq:SobolIndices order="2" total="true"/>
    </uq:DesiredResults>

    <uq:SimulationSetting stopTime="3600" interval="1.0"
        tolerance="1e-6" method="CVODE"/>

    <uq:SamplingMethod>
      <uq:LatinHyperCube numberOfSamples="1024"/>
    </uq:SamplingMethod>

    <uq:ResultSet>
      <uq:Results>
        <uq:ErrorMetric type="S1" name="S1_T_ambient" value="0.62"/>
        <uq:ErrorMetric type="ST" name="ST_T_ambient" value="0.68"/>
      </uq:Results>
    </uq:ResultSet>

  </uq:SensitivityAnalysis>
```

The `SensitivityAnalysis` element represents the standalone analysis exposed as a top-level activity. Sensitivity analysis used internally within other activities is not modeled separately.

8.4. Supporting Types

8.4.1. ReferenceDataType

Container for benchmark, analytical, or experimental data sources used in Verification and Validation:

```
<xs:complexType name="ReferenceDataType">
  <xs:sequence>
    <xs:element name="DataSource" minOccurs="0" maxOccurs="unbounded">
      <xs:complexType>
        <xs:attribute name="name" type="xs:string" use="required"/>
        <xs:attribute name="file" type="xs:string" use="optional"/>
        <xs:attribute name="type" type="xs:string" use="optional"/>
```

```
          <xs:attribute name="description" type="xs:string" use="optional"/>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
    <xs:attribute name="description" type="xs:string" use="optional"/>
  </xs:complexType>
```

8.4.2. ResultSetType / ResultsType

Unified result container for all activity types. Each `Results` element represents a single evaluation point and can contain distribution elements (for Forward UQ output) or `ErrorMetric` elements (for V&V output). Note: `ErrorMetric` does not carry a `unit` attribute — units are declared once on the corresponding `ObservedVariable`:

```
  <xs:complexType name="ResultsType">
    <xs:choice minOccurs="0" maxOccurs="unbounded">
      <xs:group ref="uq:GDistribution"/>
      <xs:element name="ErrorMetric">
        <xs:complexType>
          <xs:attribute name="name" type="xs:string" use="optional"/>
          <xs:attribute name="type" type="xs:string" use="required"/>
          <xs:attribute name="value" type="xs:double" use="required"/>
          <xs:attribute name="threshold" type="xs:double" use="optional"/>
          <xs:attribute name="pass" type="xs:boolean" use="optional"/>
        </xs:complexType>
      </xs:element>
    </xs:choice>
    <xs:attribute name="point" type="xs:string" use="optional"/>
    <xs:attributeGroup ref="uq:AExternalSource"/>
  </xs:complexType>

  <xs:complexType name="ResultSetType">
    <xs:sequence>
      <xs:element name="Results" type="uq:ResultsType"
                  minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>
```

8.5. Complete Multi-Activity Example

```
<?xml version="1.0" encoding="UTF-8"?>
<!-- Example: Multiple activity types in Activities.
     Note: activityDomainRef values reference ActivityDomain elements that would
     be defined in a Domains section (see example_domains_only.xml or
     example_instance.xml for complete domain definitions). -->
<uq:UncertaintyQuantification
    xmlns:uq="http://openscaling.org/UQ1/UncertaintyQuantification"
```

```
    name="MultiActivityDemo">

  <uq:Model name="ThermalModel" file="model.fmu"/>

  <uq:Activities id="es_01" credibilityLevel="CL4">

    <!-- tag::Verification[] -->
    <uq:Verification id="verif_01"
        name="Grid Convergence Study"
        activityDomainRef="do_verification">

      <uq:SimulationSetting stopTime="3600" interval="1.0"
          tolerance="1e-8" method="CVODE"/>

      <uq:ReferenceData description="Analytical solution">
        <uq:DataSource name="analytical_ref"
            file="reference/analytical_solution.csv"
            type="text/csv"/>
      </uq:ReferenceData>

      <uq:ResultSet>
        <uq:Results>
          <uq:ErrorMetric type="GCI" name="GCI_fine" value="0.012"
              threshold="0.05" pass="true"/>
          <uq:ErrorMetric type="observed_order" name="observed_order" value="1.95"
              threshold="1.8" pass="true"/>
        </uq:Results>
      </uq:ResultSet>

    </uq:Verification>
    <!-- end::Verification[] -->

    <!-- tag::Calibration[] -->
    <uq:Calibration id="cal_01"
        name="Convection Coefficient Calibration"
        activityDomainRef="do_calibration">

      <uq:ParameterSet>
        <uq:Parameters>
          <uq:UncertainParameter name="h_convection" source="Estimated">
            <uq:Uniform minimum="10" maximum="50" unit="W/(m2.K)"/>
          </uq:UncertainParameter>
        </uq:Parameters>
      </uq:ParameterSet>

      <uq:SimulationSetting stopTime="1800" interval="1.0"
          tolerance="1e-6" method="CVODE"/>

      <uq:ReferenceData description="Lab measurements">
        <uq:DataSource name="lab_data_2024"
            file="reference/lab_convection.csv"
```

```
            type="text/csv"/>
      </uq:ReferenceData>

      <uq:CalibrationTargets>
        <uq:Target parameterName="h_convection"
            initialValue="25.0"
            lowerBound="10.0"
            upperBound="50.0"
            unit="W/(m2.K)"/>
      </uq:CalibrationTargets>

      <uq:ResultSet>
        <uq:Results>
          <uq:Normal mu="23.7" sigma="1.2"/>
        </uq:Results>
      </uq:ResultSet>

    </uq:Calibration>
    <!-- end::Calibration[] -->

    <!-- tag::Validation[] -->
    <uq:Validation id="val_01"
        name="Wind Tunnel Comparison"
        activityDomainRef="do_validation">

      <uq:ParameterSet>
        <uq:Parameters>
          <uq:UncertainParameter name="T_ambient" source="Measured">
            <uq:Normal mu="22.0" sigma="1.0" unit="degC"/>
          </uq:UncertainParameter>
        </uq:Parameters>
      </uq:ParameterSet>

      <uq:SimulationSetting stopTime="1800" interval="0.5"
          tolerance="1e-6" method="CVODE"/>

      <uq:SamplingMethod>
        <uq:LatinHyperCube numberOfSamples="200"/>
      </uq:SamplingMethod>

      <uq:ReferenceData description="Wind tunnel data">
        <uq:DataSource name="wind_tunnel_2024"
            file="reference/wind_tunnel.csv"
            type="text/csv"/>
      </uq:ReferenceData>

      <uq:ResultSet>
        <uq:Results>
          <uq:ErrorMetric type="RMSE" name="RMSE" value="2.3"
              threshold="5.0" pass="true"/>
          <uq:ErrorMetric type="validationMetric" name="validation_metric"
```

```
value="0.87"
              threshold="0.80" pass="true"/>
        </uq:Results>
      </uq:ResultSet>

    </uq:Validation>
    <!-- end::Validation[] -->

    <!-- tag::SensitivityAnalysis[] -->
    <uq:SensitivityAnalysis id="sa_01"
        name="Global Sensitivity (Sobol)">

      <uq:ParameterSet>
        <uq:Parameters>
          <uq:UncertainParameter name="T_ambient" source="Measured">
            <uq:Uniform minimum="-10" maximum="45" unit="degC"/>
          </uq:UncertainParameter>
          <uq:UncertainParameter name="v_vehicle" source="Estimated">
            <uq:Uniform minimum="0" maximum="200" unit="km/h"/>
          </uq:UncertainParameter>
          <uq:UncertainParameter name="h_convection" source="Calibrated">
            <uq:Uniform minimum="15" maximum="35" unit="W/(m2.K)"/>
          </uq:UncertainParameter>
        </uq:Parameters>
      </uq:ParameterSet>

      <uq:ObservedVariables>
        <uq:ObservedVariable name="T_coolant_max"/>
      </uq:ObservedVariables>

      <uq:DesiredResults>
        <uq:SobolIndices order="2" total="true"/>
      </uq:DesiredResults>

      <uq:SimulationSetting stopTime="3600" interval="1.0"
          tolerance="1e-6" method="CVODE"/>

      <uq:SamplingMethod>
        <uq:LatinHyperCube numberOfSamples="1024"/>
      </uq:SamplingMethod>

      <uq:ResultSet>
        <uq:Results>
          <uq:ErrorMetric type="S1" name="S1_T_ambient" value="0.62"/>
          <uq:ErrorMetric type="ST" name="ST_T_ambient" value="0.68"/>
        </uq:Results>
      </uq:ResultSet>

    </uq:SensitivityAnalysis>
    <!-- end::SensitivityAnalysis[] -->
```

```
    <!-- ============================================================ -->
    <!-- Forward UQ (existing, unchanged)                             -->
    <!-- ============================================================ -->
    <uq:ForwardUncertaintyQuantification id="fuq_01"
        activityDomainRef="do_uq">
      <!-- ... standard ForwardUQ content ... -->
    </uq:ForwardUncertaintyQuantification>

  </uq:Activities>

</uq:UncertaintyQuantification>
```

9. Integration with SSP Traceability

9.1. Motivation

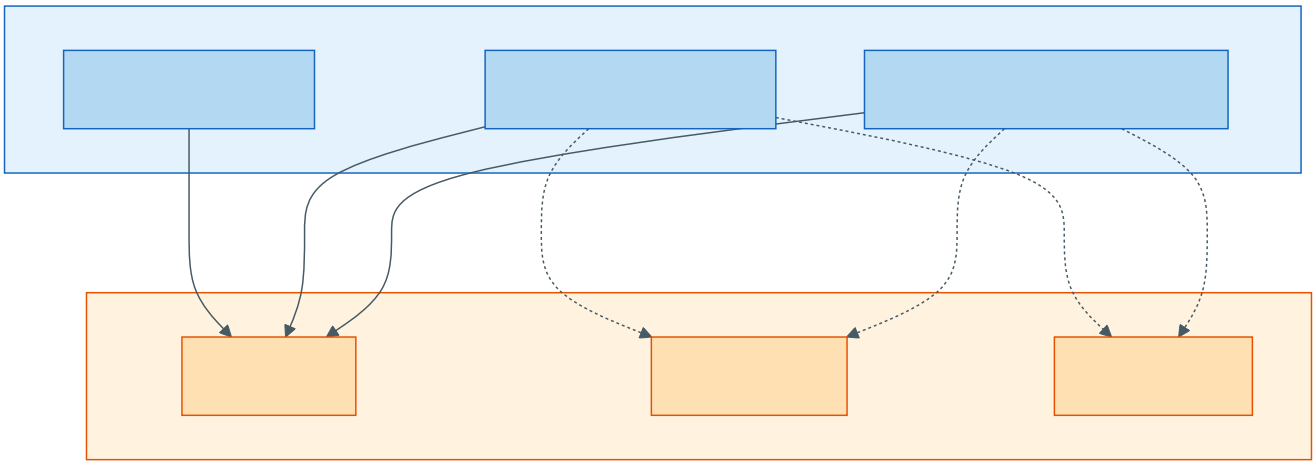
Some organizations using LS-UQ will also use (or plan to use) the **SSP Traceability** standard, which defines GlueParticle workflows for managing simulation tasks (STMD) and decision tasks (DTMD), plus resource metadata files (SRMD). The reverse also applies: organizations already using SSP Traceability may want to adopt LS-UQ. Rather than redefining concepts that SSP Traceability already covers (classifications, process phases, resource metadata), LS-UQ imports the **SSP Traceability Common** schema (`STC.xsd`) and reuses its types directly.

This design decision (see Design Decisions #9 and #10) ensures that LS-UQ data is compatible with SSP Traceability tooling while remaining functional as a standalone standard.

9.2. Architecture

LS-UQ integrates with SSP Traceability through two mechanisms:

1. **Schema import**: All three LS-UQ schema files import `STC.xsd` and reference the `stc:GElementCommon` group, which provides optional `Classification` (0..*) and `Annotations` (0..1) child elements on key types. This gives LS-UQ keyword-value metadata without having to define its own classification system.
2. **URI reference attributes**: Optional `xs:anyURI` attributes (`stmdRef`, `srmdRef`, `glueParticleRef`) on selected types allow LS-UQ elements to link to external SSP Traceability files (STMD/DTMD GlueParticles and SRMD resource metadata). These links have no effect when SSP Traceability is not used.



- **Solid arrows**: schema imports — must be present at validation time
- **Dashed arrows**: optional URI references — LS-UQ documents are valid without them

9.3. Three Usage Scenarios

LS-UQ is designed to work at three levels of SSP Traceability adoption:

Scenario	Situation	What LS-UQ Uses	Benefit
A	LS-UQ only (standalone)	`stc:Classification` elements for structured metadata. URI attributes (`stmdRef`, `srmdRef`, `glueParticleRef`) are left empty or omitted.	LS-UQ functionality without SSP Traceability workflow overhead. STC types provide more structure than plain strings at no extra cost.
B	LS-UQ first, adopt SSP Traceability later	Initially same as Scenario A. When SSP Traceability is adopted, populate URI attributes to link to STMD/DTMD GlueParticle files and SRMD resource metadata.	Existing LS-UQ data already uses STC types, so classifications are directly reusable. Adding URI references is additive; no data migration required.
C	SSP Traceability already adopted, add LS-UQ	LS-UQ references existing STMD/SRMD files via URI attributes. `stc:Classification` values are consistent with existing GlueParticle metadata.	LS-UQ uses the existing traceability infrastructure. Process context links to STMD phases via `glueParticleRef`.

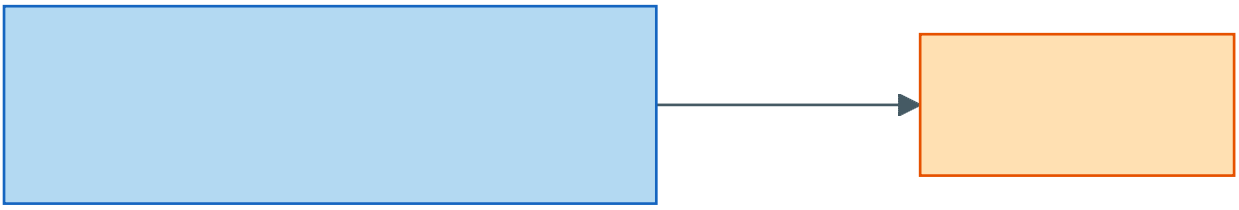


Figure 3. Scenario A — Standalone

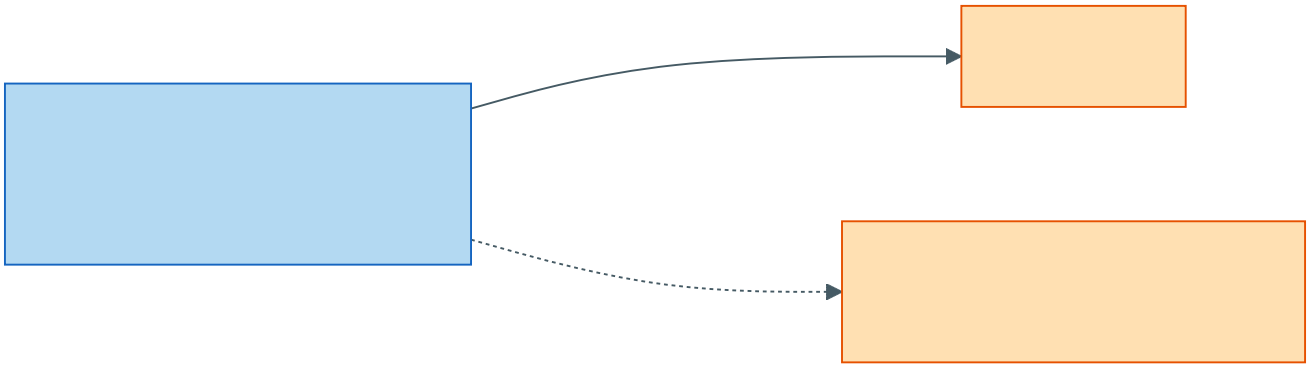


Figure 4. Scenario B — LS-UQ first, add SSP Traceability

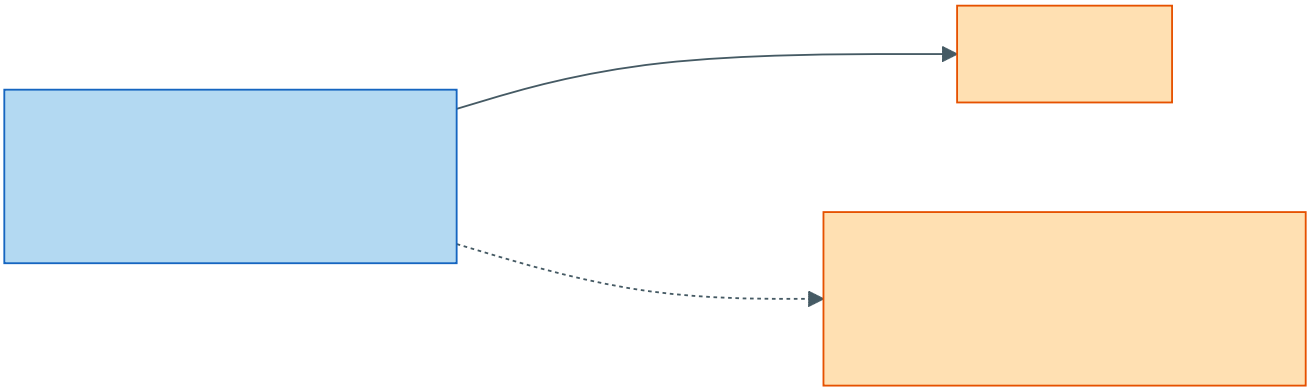


Figure 5. Scenario C — SSP Traceability first, add LS-UQ

9.4. STC Integration Points in the Schema

All types below reference `stc:GElementCommon`, which provides `Classification` (0..*) and `Annotations` (0..1) child elements. Some additionally carry URI reference attributes for linking to GlueParticle or resource metadata files:

Schema File	Type	STC Integration	Purpose
UncertaintyQuantification.xsd	`UncertaintyQuantification` (root)	`stc:GElementCommon` + `stmdRef` (xs:anyURI) + `srmdRef` (xs:anyURI)	Classify the overall UQ document. Link to the STMD describing the simulation task and the SRMD describing the simulation resource.
UncertaintyQuantification.xsd	`ActivitiesType`	`stc:GElementCommon`	Classify the activities container (e.g., by credibility framework or organizational context).
DomainHierarchy.xsd	`CredibilityAssessmentType`	`stc:GElementCommon`	Richer credibility metadata via keyword-value pairs, alongside the simple `level` /`purpose`/`standard` string attributes.
DomainHierarchy.xsd	`ProcessContextType`	`stc:GElementCommon` + `glueParticleRef` (xs:anyURI)	Classify the process context. Link directly to an STMD/DTMD GlueParticle for process phase traceability.
DomainHierarchy.xsd	`TypedOperationalDomainType`	`stc:GElementCommon` + `srmdRef` (xs:anyURI)	Classify a domain. Link to an SRMD describing this domain's resource metadata.

Schema File	Type	STC Integration	Purpose
DomainHierarchy.xsd	`ActivityDomainType`	`stc:GElementCommon` + `srmdRef` (xs:anyURI)	Classify an activity domain. Link to an SRMD for the activity-specific domain.
Assumptions.xsd	`AssumptionType`	`stc:GElementCommon`	Classify individual assumptions (e.g., by source, confidence level, or organizational category).

9.5. ProcessContextType: Bridging Process Phases

`ProcessContextType` bridges LS-UQ's domain hierarchy with SSP Traceability's process phase definitions.

The SSP Traceability standard defines process phases authoritatively in `STMD.xsd` (7 phases for the Credible Simulation Process) and `DTMD.xsd` (5 phases for the Credible Decision Process). Each phase contains named steps (e.g., `AnalyzeSimulationTaskAndObjectives`, `ImplementModel`, `DecideSimulationObjectiveFulfillment` in STMD; `AnalyzeDecisionTask`, `DefineSubTasks` in DTMD). Rather than reduplicate these definitions, LS-UQ's `ProcessContextType` uses:

- `process` attribute (xs:string): lightweight indicator: `CDP`, `CSP`, or `CMP`. CDP and CSP map to DTMD and STMD GlueParticle files respectively; CMP (Credible Modeling Process) is an OpenSCALING concept without a dedicated SSP Traceability file type.
- `phase` attribute (xs:string): phase name matching STMD/DTMD definitions
- `glueParticleRef` attribute (xs:anyURI): direct link to the STMD/DTMD GlueParticle

In **Scenario A** (standalone), only the `process` and `phase` string attributes are needed:

```
<uq:ProcessContext process="CSP" phase="ImplementModel"
                   description="Thermal model implementation"/>
```

In **Scenario B or C** (with SSP Traceability), `glueParticleRef` provides a machine-resolvable link:

```
<uq:ProcessContext process="CSP" phase="ImplementModel"
                   glueParticleRef="metadata/thermal_analysis.stmd#ImplementModel"/>
```

Both forms are valid simultaneously. The string attributes are human-readable; the URI is machine-resolvable.

9.6. Upgrade Path: Scenario A to B

Organizations starting with Scenario A can adopt SSP Traceability incrementally:

1. **No changes needed** to existing LS-UQ documents; they already use STC types via the schema

import.

2. **Populate URI attributes** on relevant elements (`stmdRef`, `srmdRef`, `glueParticleRef`) as STMD/DTMD GlueParticle files and SRMD metadata files are created.
3. **Enrich `stc:Classification` elements** with keywords matching the organization's GlueParticle metadata conventions.

No data migration is required.

9.7. Upgrade Path: Scenario B to C

Organizations that started with LS-UQ and incrementally adopted SSP Traceability (Scenario B) can move to full SSP Traceability integration (Scenario C) by shifting the source of authority:

1. **Make SSP Traceability the primary metadata source.** Instead of maintaining `stc:Classification` values independently in LS-UQ documents, derive them from the organization's GlueParticle and SRMD conventions. LS-UQ classifications become consistent with — or generated from — the central traceability metadata.
2. **Ensure all URI attributes are populated.** Every `stmdRef` and `glueParticleRef` should resolve to an actual GlueParticle file; every `srmdRef` should resolve to an SRMD resource metadata file. In Scenario B these may have been partially filled; in Scenario C they are expected to be complete.
3. **Use `ProcessContext.glueParticleRef` as the authoritative process link.** The `process` and `phase` string attributes remain as human-readable summaries, but tooling should resolve traceability through the GlueParticle URI.

No schema changes are required.

10. Changes to Existing Types

All modifications to existing types are **additive**: new optional attributes or optional child elements only.

10.1. UncertaintyQuantification (Root Element)

Change	Before	After
New children	`Model`, `Activities`	`RequiredAssumptions` (0..*), `Domains` (0..1), `Model` (0..1), `Activities`, `stc:GElementCommon` (Classification + Annotations)
New attributes	-	`requiredAssumptionsRefs` (xs:string, optional), `stmdRef` (xs:anyURI, optional), `srmdRef` (xs:anyURI, optional)

10.2. ModelType

Change	Before	After
Structure	Attributes only (no children)	Optional `ModelingAssumptions` (0..*) child
Attributes	`name`, `file`, `model`, `modelResourceMetaData`	Unchanged

10.3. ActivitiesType

Change	Before	After
ForwardUQ	Required (1..1)	Optional (0..*)
New children	-	`Verification` (0..**), `Calibration` (0..**), `Validation` (0..**), `SensitivityAnalysis` (0..**), `stc:GElementCommon` (Classification + Annotations)
New attribute	`id` (xs:ID)	`id` (xs:ID), `credibilityLevel` (xs:string, optional)

10.4. ForwardUncertaintyQuantificationType

Change	Before	After
New attribute	-	`activityDomainRef` (xs:string, optional)

Change	Before	After
Children	All required (minOccurs=1)	All optional (minOccurs=0), consistent with other activity types

10.5. OperationalDomainType

Change	Before	After
New attributes	-	`id` (xs:ID, optional), `name` (xs:string, optional), `purpose` (xs:string, optional)
Children	Unchanged	Unchanged

10.6. AExternalSource Attribute Group

Elements like `Parameters`, `Axes`, `ConvexHull`, `HyperRectangle`, `Points`, `Units`, `ObservedVariables`, `DesiredResults`, `SimulationSetting`, and `SamplingMethod` can grow very large or reference external configuration files. To avoid bloating the XML document, these types now support externalizing their data to files (CSV, XML, etc.) following SSP/FMI conventions.

The `AExternalSource` attribute group is a reusable set of optional attributes:

```
  <xs:attributeGroup name="AExternalSource">
    <xs:attribute name="source"       type="xs:anyURI"          use="optional"/>
    <xs:attribute name="type"         type="xs:normalizedString" use="optional"/>
    <xs:attribute name="sourceBase"   type="xs:string"          use="optional"
default="file"/>
    <xs:attribute name="checksum"     type="xs:string"          use="optional"/>
    <xs:attribute name="checksumType" type="xs:string"          use="optional"/>
  </xs:attributeGroup>
```

Attribute	Type	Purpose
`source`	`xs:anyURI`	URI of the external data file (e.g., `resources/lhs_10k.csv`)
`type`	`xs:normalizedString`	MIME type of the external file (e.g., `text/csv`, `application/xml`)
`sourceBase`	`xs:string` (default: `file`)	Base for resolving relative URIs, following SSP conventions
`checksum`	`xs:string`	Integrity checksum of the external file
`checksumType`	`xs:string`	Algorithm used for the checksum (e.g., `SHA-256`)

When `source` is present, inline children may be omitted. This is enabled by relaxing `minOccurs` to `0` on the child elements of the affected types.

10.6.1. Where You Can Use It

The following table lists every element in the schema that carries the `AExternalSource` attributes, with a small XML snippet showing how to use it:

Element	Parent	Example

`Parameters`	`ParameterSet`	`<uq:ParameterSet description="500 calibrated parameters">`
		`  <uq:Parameters`
		`  source="resources/params.xml"`
		`  type="application/xml"/>`
		`</uq:ParameterSet>`

`Axes`	`OperationalDomain`	`<uq:OperationalDomain>`
		`  <uq:Axes source="resources/axes.xml"`
		`  type="application/xml"/>`
		`  ...`
		`</uq:OperationalDomain>`

`ConvexHull`	`Boundary`	`<uq:Boundary>`
		`  <uq:ConvexHull`
		`  source="resources/hull_vertices.csv"`
		`  type="text/csv"/>`
		`</uq:Boundary>`

`HyperRectangle`	`Boundary`	`<uq:Boundary>`
		`  <uq:HyperRectangle`
		`  source="resources/bounds.xml"`
		`  type="application/xml"/>`
		`</uq:Boundary>`

`Points`	`PointSet`	`<uq:PointSet id="lhs_10k">`
		`  <uq:Points source="resources/lhs_10k.csv"`
		`  type="text/csv"`
		`  checksum="a1b2c3..." checksumType="SHA-256"/>`
		`  <uq:Parameters>...</uq:Parameters>`
		`</uq:PointSet>`

`Units`	`ParameterSet`	`<uq:ParameterSet>`
		`  ...`
		`  <uq:Units source="resources/units.xml"`
		`  type="application/xml"/>`
		`</uq:ParameterSet>`

Element	Parent	Example

`ObservedVariables`	Activity types (Verification, Validation, etc.)	`<uq:Validation id="val_01">`
		`  <uq:ObservedVariables`
		`  source="resources/observed_vars.xml"`
		`  type="application/xml"/>`
		`  ...`
		`</uq:Validation>`

`DesiredResults`	Activity types (Verification, Validation, SA, Forward UQ)	`<uq:ForwardUncertaintyQuantification id="fuq_01">`
		`  <uq:DesiredResults`
		`  source="resources/desired_results.xml"`
		`  type="application/xml"`
		`  description="Externalized desired results"/>`
		`  ...`
		`</uq:ForwardUncertaintyQuantification>`

`SimulationSetting`	Activity types (Verification, Calibration, Validation, SA, Forward UQ)	`<uq:ForwardUncertaintyQuantification id="fuq_01">`
		`  <uq:SimulationSetting`
		`  source="resources/sim_settings.xml"`
		`  type="application/xml"`
		`  description="Externalized simulation settings"/>`
		`  ...`
		`</uq:ForwardUncertaintyQuantification>`

`SamplingMethod`	Activity types (Verification, Validation, SA, Forward UQ)	`<uq:ForwardUncertaintyQuantification id="fuq_01">`
		`  <uq:SamplingMethod`
		`  source="resources/sampling_method.xml"`
		`  type="application/xml"`
		`  description="Externalized sampling method"/>`
		`  ...`
		`</uq:ForwardUncertaintyQuantification>`

10.6.2. Applied to ParametersType

Change	Before	After
New attributes	`id`	`id`, `source`, `type`, `sourceBase`, `checksum`, `checksumType` (via `AExternalSource`)

10.6.3. Applied to AxesType

Change	Before	After
New attributes	-	`source`, `type`, `sourceBase`, `checksum`, `checksumType` (via `AExternalSource`)

10.6.4. Applied to ConvexHullType

Change	Before	After
Type	Anonymous type inside `BoundaryType`	Named `ConvexHullType` with `AExternalSource`
New attributes	-	`source`, `type`, `sourceBase`, `checksum`, `checksumType` (via `AExternalSource`)

10.6.5. Applied to HyperRectangleType

Change	Before	After
Type	Anonymous type inside `BoundaryType`	Named `HyperRectangleType` with `AExternalSource`
New attributes	-	`source`, `type`, `sourceBase`, `checksum`, `checksumType` (via `AExternalSource`)

10.6.6. Applied to UnitsType

Change	Before	After
Type	`ssc:TUnits` (SSC schema)	`uq:UnitsType` — wrapper around `ssc:TUnit` with `AExternalSource`
New attributes	-	`source`, `type`, `sourceBase`, `checksum`, `checksumType` (via `AExternalSource`)

10.6.7. Applied to ObservedVariablesType

Change	Before	After
New attributes	-	`source`, `type`, `sourceBase`, `checksum`, `checksumType` (via `AExternalSource`)

10.6.8. Applied to ObservedVariableType

Change	Before	After
New attribute `unit`	-	Optional `xs:string`. Declares the physical unit for this observed variable. Units are declared here rather than on individual result values (`ErrorMetric`, distribution elements) inside `ResultSet`, avoiding redundancy.

10.6.9. Applied to PointsType

Change	Before	After
`Point` minOccurs	1 (implicit)	0 (source-only usage valid)
New attributes	-	`source`, `type`, `sourceBase`, `checksum`, `checksumType` (via `AExternalSource`)

10.6.10. Applied to DesiredResultsType

Change	Before	After
New attributes	-	`description`, `source`, `type`, `sourceBase`, `checksum`, `checksumType` (via `AExternalSource`)

10.6.11. Applied to SimulationSettingType

Change	Before	After
`stopTime`	`use="required"`	`use="optional"` (omitted when externalized)
`interval`	`use="required"`	`use="optional"` (omitted when externalized)
New attributes	-	`description`, `source`, `type`, `sourceBase`, `checksum`, `checksumType` (via `AExternalSource`)

10.6.12. Applied to SamplingMethodType

Change	Before	After
Choice element (PseudoRandom / LatinHyperCube / VendorSpecificSampling)	Required (minOccurs=1)	Optional (minOccurs=0, wrapped in sequence) — omitted when externalized

Change	Before	After
New attributes	-	`description`, `source`, `type`, `sourceBase`, `checksum`, `checksumType` (via `AExternalSource`)

10.6.13. Usage Examples

The external source attributes can be used on any of the ten types listed above. Below are examples showing inline vs external usage.

PointsType — external CSV with checksum

Large sample sets (e.g., 10,000 LHS points) are stored in a CSV file. No inline `Point` children are needed:

```
<uq:PointSet id="lhs_10k">
  <!-- 10k points externalized to CSV — no inline Point elements -->
  <uq:Points
      source="resources/lhs_10k.csv"
      type="text/csv"
      checksum="a1b2c3d4e5f6..."
      checksumType="SHA-256"/>
  <uq:Parameters>
    <uq:UncertainParameter name="temperature">
      <uq:Uniform minimum="-10" maximum="80"/>
    </uq:UncertainParameter>
  </uq:Parameters>
</uq:PointSet>
```

A small reference set can stay inline for readability:

```
<uq:PointSet id="reference_pts">
  <uq:Points>
    <uq:Point id="p1" coordinates="-10 0.5"/>
    <uq:Point id="p2" coordinates="25 1.5"/>
    <uq:Point id="p3" coordinates="80 3.0"/>
  </uq:Points>
  ...
</uq:PointSet>
```

ParametersType — external XML

A large set of parameters (hundreds of uncertain parameters) can be stored in a separate XML file. The `source` attribute is on the `Parameters` element, not on `ParameterSet`:

```
<uq:ForwardUncertaintyQuantification id="fuq_01">
  <uq:ParameterSet description="500 calibrated parameters">
```

```
    <!-- Parameters externalized to an XML file -->
    <uq:Parameters
        source="resources/params.xml"
        type="application/xml"/>
  </uq:ParameterSet>
  <uq:ObservedVariables>...</uq:ObservedVariables>
  <uq:DesiredResults mean="true" standardDeviation="true"/>
  <uq:SimulationSetting stopTime="100.0" interval="0.1" method="CVode"/>
  <uq:SamplingMethod>
    <uq:LatinHyperCube numberOfSamples="10000"/>
  </uq:SamplingMethod>
</uq:ForwardUncertaintyQuantification>
```

Operational domain children — external data

Individual children of `OperationalDomain` can be externalized independently. For example, axes from an external file and boundary points from a CSV:

```
<uq:OperationalDomain>
  <uq:Axes source="resources/axes.xml"
      type="application/xml"/>
  <uq:Boundary>
    <uq:HyperRectangle
        source="resources/bounds.csv"
        type="text/csv"/>
  </uq:Boundary>
  <uq:ExperimentPoints>
    <uq:PointSet id="lhs_10k">
      <uq:Points source="resources/lhs_10k.csv"
          type="text/csv"/>
      <uq:Parameters>...</uq:Parameters>
    </uq:PointSet>
  </uq:ExperimentPoints>
</uq:OperationalDomain>
```

A complete example exercising all three patterns is in `examples/example_external_source.xml`.

11. Migration Examples: Old vs Proposed Schema

The following side-by-side comparisons show how the two original LS-UQ examples (DC Motor and Mobile Valve) are expressed in the current format versus the proposed schema.

11.1. DC Motor

11.1.1. Old Format

```
Unresolved directive in proposal.adoc - include::../examples/dc_motor_experiment.xml[]
```

11.1.2. Proposed Format

```
Unresolved directive in proposal.adoc - include::examples/dc_motor_new.xml[]
```

11.1.3. Key Changes

- **Root element**: `ExperimentSetup` → `UncertaintyQuantification`; SSP1 namespaces replace SSP2; `stc` namespace added.
- **Structure**: `ExperimentSet` wrapper removed; activities live inside `Activities` container with an `id`.
- **Model element**: Optional in both old and new schemas; omitted here as in the original DC Motor example.
- **Simulation**: `SimulationMethod` → `SimulationSetting`; `solver` attribute → `method`; vendor annotations moved to `<KeyValues type="…">` children.
- **Sampling**: `LatinHypercube` → `LatinHyperCube` (capital C); `randomNumberGenerator`/`randomNumberSeed` moved into `<KeyValues>`.

11.2. Mobile Valve

11.2.1. Old Format

```
<?xml version="1.0" encoding="UTF-8"?>

<uq:ExperimentSetup
  name="ScenarioMobileValve_ForwardUQ"
  xmlns:uq="http://openscaling.org/UQ1/UncertaintyQuantification"
  xmlns:ssv="http://ssp-standard.org/SSP1/SystemStructureParameterValues"
  xmlns:ssc="http://ssp-standard.org/SSP1/SystemStructureCommon">
```

```
<uq:Model
  name="ScenarioMobileValve"
  file="ParkerSimPack.ssp"
  modelResourceMetaData="mobile_valve.srmd" />

<uq:ExperimentSet>
  <uq:ForwardUncertaintyQuantification>
    <uq:ParameterSet>
      <uq:Parameters>
        <uq:UncertainParameter name="KqMI_PABT_gain" source="Estimated"
          sourceInfo="Parker Hannifin's Hydraulic Valve Systems">
          <uq:Normal mu="1" sigma="0.1" minimum="0.8" maximum="1.2" unit="-" />
        </uq:UncertainParameter>
        <uq:UncertainParameter name="KqMO_PABT_gain" source="Estimated"
          sourceInfo="Parker Hannifin's Hydraulic Valve Systems">
          <uq:Normal mu="1" sigma="0.1" minimum="0.8" maximum="1.2" unit="-" />
        </uq:UncertainParameter>
      </uq:Parameters>

      <uq:OperationalDomain>
        <uq:Axes>
          <uq:Axis name="uA" unit="mA" />
          <uq:Axis name="pTB" unit="bar" />
        </uq:Axes>

        <uq:Boundary>
          <uq:ConvexHull>
            <uq:Point coordinates="0 0" />
            <uq:Point coordinates="0 20" />
            <uq:Point coordinates="900 20" />
            <uq:Point coordinates="900 0" />
          </uq:ConvexHull>
        </uq:Boundary>

        <uq:ExperimentPoints>
          <uq:PointSet id="PS1">
            <uq:Points>
              <uq:Point id="P1" coordinates="345 1" />
              <uq:Point id="P2" coordinates="355 1" />
              <uq:Point id="P3" coordinates="375 1" />
              <uq:Point id="P4" coordinates="395 1" />
              <uq:Point id="P5" coordinates="415 1" />
              <uq:Point id="P6" coordinates="440 1" />
              <uq:Point id="P7" coordinates="465 1" />
              <uq:Point id="P8" coordinates="485 1" />
              <uq:Point id="P9" coordinates="505 1" />
              <uq:Point id="P10" coordinates="525 1" />
            </uq:Points>
          </uq:PointSet>

        </uq:ExperimentPoints>
```

```
        </uq:OperationalDomain>

        <uq:Units>
          <ssc:Unit name="-">
            <ssc:BaseUnit factor="1" offset="0" />
          </ssc:Unit>
          <ssc:Unit name="bar">
            <ssc:BaseUnit kg="1" m="-1" s="-2" factor="1e5" offset="0.0" />
          </ssc:Unit>
          <ssc:Unit name="mA">
            <ssc:BaseUnit A="1" factor="1e-3" offset="0.0" />
          </ssc:Unit>
          <ssc:Unit name="L/min">
            <ssc:BaseUnit m="3" s="-1" factor="6e-4" />
          </ssc:Unit>
        </uq:Units>
      </uq:ParameterSet>

      <uq:ObservedVariables>
        <uq:ObservedVariable name="qA" distributionApproximation="Normal" />
        <uq:ObservedVariable name="pA" distributionApproximation="Normal" />
      </uq:ObservedVariables>

      <uq:DesiredResults
        data="mobile_valve.csv"
        summary="mobile_valve_result_summary.xml"
        scope="FinalTime"
        histogram="true">
      </uq:DesiredResults>

      <!-- TODO: Viktor -->
      <uq:SimulationMethod stopTime="0.5" interval="0.001" tolerance="1e-4"
solver="FixedStepMaster" />

      <uq:SamplingMethod>
        <uq:PseudoRandom numberOfSamples="1000" />
      </uq:SamplingMethod>
    </uq:ForwardUncertaintyQuantification>
  </uq:ExperimentSet>
</uq:ExperimentSetup>
```

11.2.2. Proposed Format

```
<?xml version="1.0" encoding="UTF-8"?>
<!--
  Mobile Valve — Proposed Schema Format
  Migrated from examples/mobile_valve/mobile_valve_setup.xml
-->
<uq:UncertaintyQuantification
```

```
    name="ScenarioMobileValve_ForwardUQ"
    xmlns:uq="http://openscaling.org/UQ1/UncertaintyQuantification"
    xmlns:ssv="http://ssp-standard.org/SSP1/SystemStructureParameterValues"
    xmlns:ssc="http://ssp-standard.org/SSP1/SystemStructureCommon"
    xmlns:stc="http://ssp-standard.org/SSPTraceability1/SSPTraceabilityCommon">

  <uq:Model
      name="ScenarioMobileValve"
      file="ParkerSimPack.ssp"
      modelResourceMetaData="mobile_valve.srmd"/>

  <uq:Activities id="act_01">
    <uq:ForwardUncertaintyQuantification id="fuq_01">

      <uq:ParameterSet>
        <uq:Parameters>
          <uq:UncertainParameter name="KqMI_PABT_gain" source="Estimated"
              sourceInfo="Parker Hannifin's Hydraulic Valve Systems">
            <uq:Normal mu="1" sigma="0.1" minimum="0.8" maximum="1.2" unit="-"/>
          </uq:UncertainParameter>
          <uq:UncertainParameter name="KqMO_PABT_gain" source="Estimated"
              sourceInfo="Parker Hannifin's Hydraulic Valve Systems">
            <uq:Normal mu="1" sigma="0.1" minimum="0.8" maximum="1.2" unit="-"/>
          </uq:UncertainParameter>
        </uq:Parameters>

        <uq:OperationalDomain>
          <uq:Axes>
            <uq:Axis name="uA" unit="mA"/>
            <uq:Axis name="pTB" unit="bar"/>
          </uq:Axes>

          <uq:Boundary>
            <uq:ConvexHull>
              <uq:Point coordinates="0 0"/>
              <uq:Point coordinates="0 20"/>
              <uq:Point coordinates="900 20"/>
              <uq:Point coordinates="900 0"/>
            </uq:ConvexHull>
          </uq:Boundary>

          <uq:ExperimentPoints>
            <uq:PointSet id="PS1">
              <uq:Points>
                <uq:Point id="P1" coordinates="345 1"/>
                <uq:Point id="P2" coordinates="355 1"/>
                <uq:Point id="P3" coordinates="375 1"/>
                <uq:Point id="P4" coordinates="395 1"/>
                <uq:Point id="P5" coordinates="415 1"/>
                <uq:Point id="P6" coordinates="440 1"/>
                <uq:Point id="P7" coordinates="465 1"/>
```

```
              <uq:Point id="P8" coordinates="485 1"/>
              <uq:Point id="P9" coordinates="505 1"/>
              <uq:Point id="P10" coordinates="525 1"/>
            </uq:Points>
            <uq:Parameters>
              <uq:UncertainParameter name="KqMI_PABT_gain" source="Estimated">
                <uq:Normal mu="1" sigma="0.1" minimum="0.8" maximum="1.2" unit="-"/>
              </uq:UncertainParameter>
            </uq:Parameters>
          </uq:PointSet>
        </uq:ExperimentPoints>
      </uq:OperationalDomain>

      <uq:Units>
        <uq:Unit name="-">
          <ssc:BaseUnit factor="1" offset="0"/>
        </uq:Unit>
        <uq:Unit name="bar">
          <ssc:BaseUnit kg="1" m="-1" s="-2" factor="1e5" offset="0.0"/>
        </uq:Unit>
        <uq:Unit name="mA">
          <ssc:BaseUnit A="1" factor="1e-3" offset="0.0"/>
        </uq:Unit>
        <uq:Unit name="L/min">
          <ssc:BaseUnit m="3" s="-1" factor="6e-4"/>
        </uq:Unit>
      </uq:Units>
    </uq:ParameterSet>

    <uq:ObservedVariables>
      <uq:ObservedVariable name="qA" distributionApproximation="Normal"/>
      <uq:ObservedVariable name="pA" distributionApproximation="Normal"/>
    </uq:ObservedVariables>

    <uq:DesiredResults
        data="mobile_valve.csv"
        summary="mobile_valve_result_summary.xml"
        scope="FinalTime"
        histogram="true">
    </uq:DesiredResults>

    <uq:SimulationSetting stopTime="0.5" interval="0.001"
                          tolerance="1e-4" method="FixedStepMaster"/>

    <uq:SamplingMethod>
      <uq:PseudoRandom numberOfSamples="1000"/>
    </uq:SamplingMethod>

  </uq:ForwardUncertaintyQuantification>
</uq:Activities>
```

```
</uq:UncertaintyQuantification>
```

11.2.3. Key Changes

- **Root element**: `ExperimentSetup` → `UncertaintyQuantification`; `stc` namespace added.
- **Structure**: `ExperimentSet` wrapper removed; activities live inside `Activities` container with an `id`.
- **Simulation**: `SimulationMethod` → `SimulationSetting`; `solver` attribute → `method`.
- **PointSet**: Added `Parameters` child inside `PointSet` (required by new schema's `ExperimentPointsType`).

12. Appendix

12.1. Full Schema Files

The complete schema files are provided in the `schema/proposed/` directory:

- `schema/proposed/UncertaintyQuantification.xsd`: Main schema (modified)
- `schema/proposed/UncertainParameterType.xsd`: Distribution types (GDistribution group extracted)
- `schema/proposed/Assumptions.xsd`: Assumptions framework (new)
- `schema/proposed/DomainHierarchy.xsd`: Domain hierarchy (new)

The complete example files are in the `examples/` directory:

- `examples/example_instance.xml`: Full example with all features
- `examples/example_assumptions_only.xml`: Assumptions focus
- `examples/example_domains_only.xml`: Domain hierarchy focus
- `examples/example_activities.xml`: Activity types focus
- `examples/example_external_source.xml`: External source attributes (inline vs external data)

12.2. New Type Inventory

12.2.1. New Types in Assumptions.xsd

Type Name	Kind	Purpose
`OriginType`	simpleType	Enumeration: requirement, standard, design-decision, inherited, expert-judgment, empirical, literature, unknown
`RealValueType`	complexType	Single real value with unit
`RealRangeType`	complexType	Low/high range with unit
`StringValueType`	complexType	Short string value
`ExternalDocumentType`	complexType	Reference to external file or URL
`OperationalDomainRefType`	complexType	Reference to a domain by ID
`AssumptionType`	complexType	Core assumption with polymorphic value
`RequiredAssumptionsType`	complexType	Container for pre-experiment requirements

Type Name	Kind	Purpose
`ModelingAssumptionsType`	complexType	Container for realized model assumptions

12.2.2. New Types in DomainHierarchy.xsd

Type Name	Kind	Purpose
`TypedOperationalDomainKind`	simpleType	Enumeration: OperationalDesignDomain, RequestedOperationalDomain, ModeledOperationalDomain
`CredibilityAssessmentType`	complexType	Links credibility level to simulation purpose. Attributes: `level`, `purpose`, `standard` (all xs:string)
`ProcessContextType`	complexType	Process + phase (xs:string) + `glueParticleRef` (xs:anyURI) + `stc:GElementCommon`
`TypedOperationalDomainType`	complexType	Domain with kind, hierarchy, credibility
`ActivityDomainType`	complexType	Activity-scoped domain
`ActivitySettingsType`	complexType	Activity settings
`CoverageClassificationType`	simpleType	Enumeration: Covered, HighRisk, AcceptableRisk, NotNeeded
`DomainCoverageType`	complexType	Gap analysis between two domains
`DomainsType`	complexType	Top-level container

Several attributes in DomainHierarchy.xsd use `xs:string` instead of fixed enumerations to support different frameworks and terminologies. Recommended values are documented in the schema's XML comments. See Design Decision #8.

12.2.3. New Types in UncertaintyQuantification.xsd

Type Name	Kind	Purpose
`AExternalSource`	attributeGroup	Reusable attributes for externalizing large data to files (`source`, `type`, `sourceBase`, `checksum`, `checksumType`). Applied to `ParametersType`, `AxesType`, `ConvexHullType`, `HyperRectangleType`, `PointsType`, `UnitsType`, `ObservedVariablesType`, `DesiredResultsType`, `SimulationSettingType`, `SamplingMethodType`.
`ConvexHullType`	complexType	Convex hull boundary with `AExternalSource` (extracted from anonymous type in `BoundaryType`)

Type Name	Kind	Purpose
`HyperRectangleType`	complexType	Axis-aligned bounding box with `AExternalSource` (extracted from anonymous type in `BoundaryType`)
`UnitsType`	complexType	Wrapper around `ssc:TUnit` with `AExternalSource`; replaces `ssc:TUnits` in `ParameterSetType`
`VerificationType`	complexType	Verification activity
`ValidationType`	complexType	Validation activity
`CalibrationType`	complexType	Calibration activity with targets
`SensitivityAnalysisType`	complexType	Sensitivity analysis activity
`ReferenceDataType`	complexType	Reference data container
`ResultsType`	complexType	Per-point results (distributions or error metrics) with optional external source. `ErrorMetric` has no `unit` — units belong on `ObservedVariable`
`ResultSetType`	complexType	Container for Results elements across all evaluation points